

Mathematics

Stage 9

Paper 1

2025

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Here are some numbers written in standard form.

6.7×10^4

8.1×10^{-5}

3.2×10^6

4.8×10^{-2}

5.0×10^{-8}

Write down the largest and the smallest of these numbers.

Largest =

Smallest =

[2]

- 2 Naomi buys a mobile phone.

Draw a ring around the most appropriate value for the amount of internal storage on her mobile phone.

128 nm

128 GB

128 tonnes

128 μl

[1]

- 3 Here is an equation.

$$8^p \times 8^q = 8^2$$

Find two pairs of values of p and q to make the equation correct.

$p =$ and $q =$

$p =$ and $q =$

[2]

- 4 (a) Here is a sequence with n th term $n^2 + 4$

5, 8, 13, 20, 29, ...

Find the 12th term of this sequence.

..... [1]

- (b) Here is a different sequence.

7, 10, 15, 22, 31, ...

By comparing this sequence to the sequence in part (a), find an expression for the n th term of this sequence.

..... [1]

- 5 Draw a ring around each of the **two** irrational numbers.

3.4 $\sqrt{5}$ $\sqrt{16}$ π 12 $\frac{1}{3}$

[1]

- 6 Polygon A and polygon B are both regular.
The table gives information about each polygon.

Polygon	Interior angle	Exterior angle
A	120°	 $^\circ$
B	 $^\circ$	72°

- (a) Complete the two missing angles in the table. [1]

- (b) Explain why polygon A tessellates.

.....
..... [1]

7 Here is a formula.

$$a = b^2 - 4$$

Rearrange the formula to make b the subject.

$$b = \dots\dots\dots [2]$$

8 Line L has the equation $2y = -5x + 6$

Draw a ring around the coordinates of the y -intercept of line L.

$(0, 6)$

$(0, 2)$

$(0, 3)$

$(0, -5)$

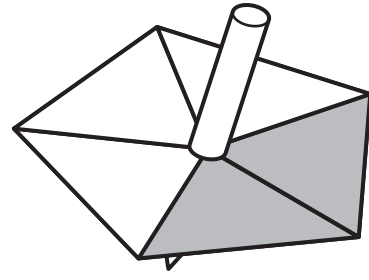
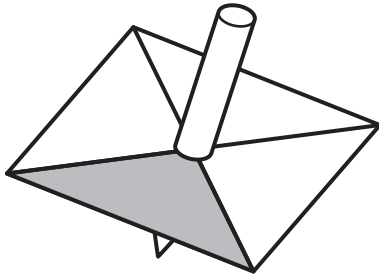
[1]

9 Tick (✓) to show if each statement is true or false.

Statement	True	False
The square root of 64 is 8 or -8		
$(-3)^2 = -9$		

[1]

- 10 Rajiv has a fair 4-sided spinner with one section shaded.
He also has a fair 5-sided spinner with two sections shaded.

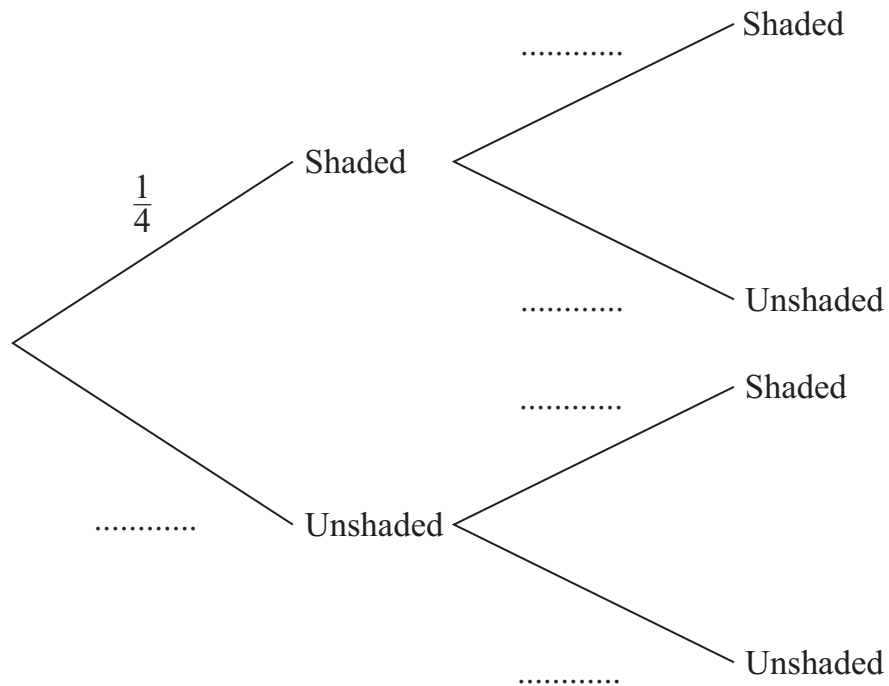


Rajiv spins each of his spinners once.

- (a) Complete the tree diagram with the five missing probabilities.

4-sided spinner

5-sided spinner



[2]

- (b) Find the probability that **both** spinners land on an unshaded section.

..... [2]

- 11** A rectangle has a perimeter of 15 cm.
The rectangle is enlarged by a scale factor of 2

(a) Work out the **perimeter** of the enlarged rectangle.

..... cm [1]

(b) The area of the rectangle is 11 cm^2 .

Work out the **area** of the enlarged rectangle.

..... cm^2 [1]

- 12** Complete the missing fraction by writing an integer in each box.

$$-\frac{6}{13} < -\frac{\boxed{}}{\boxed{}} < -\frac{5}{13}$$

[1]

13 A bag contains some counters.

Each counter is numbered 1, 2, 3 or 4

A counter is taken at random from the bag.

The table shows some of the probabilities of taking a counter with each number.

	1	2	3	4
Probability	0.1	0.25	0.3	

(a) Complete the table to show the probability of taking a counter numbered 4

[1]

(b) Mike takes a counter at random from the bag.

He records the number on the counter and replaces the counter in the bag.

He repeats this process until he has recorded 40 numbers.

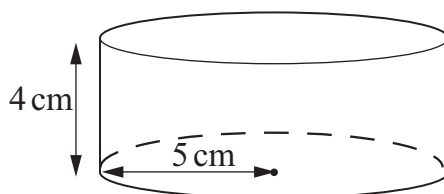
Work out how many times he should expect to record the number 3

..... [1]

(c) Find the **smallest** possible number of counters in the bag.

..... [1]

14 A cylinder has a base radius of 5 cm and a height of 4 cm.



NOT TO
SCALE

Draw a ring around the closest estimate for the volume of the cylinder.

63 cm³

79 cm³

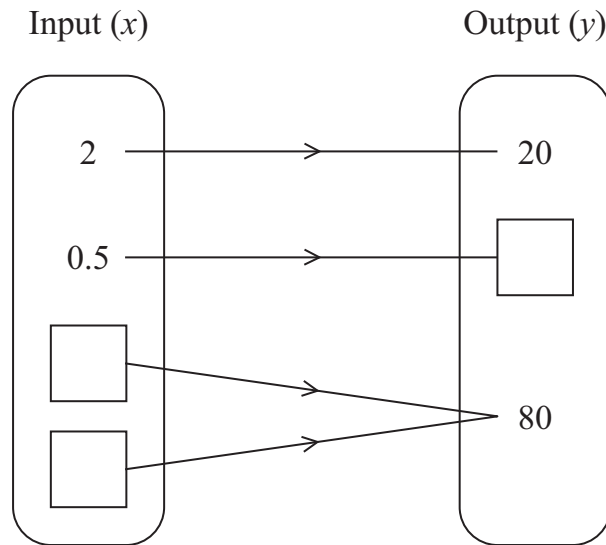
100 cm³

314 cm³

628 cm³

[1]

- 15 The mapping diagram represents the function $y = 5x^2$



Complete the diagram by writing a different number in each box.

[2]

- 16 A prism has 7 planes of symmetry.
The prism has a cross-section that is a regular polygon.

Write down the name of the regular polygon.

..... [1]

17 Solve the simultaneous equations.

$$\begin{aligned}4x - y &= 11 \\ 2x + 3y &= 2\end{aligned}$$

$$x = \text{.....}$$

$$y = \text{.....}$$

[3]

18 The cost of hiring a taxi is

- \$2 fixed cost

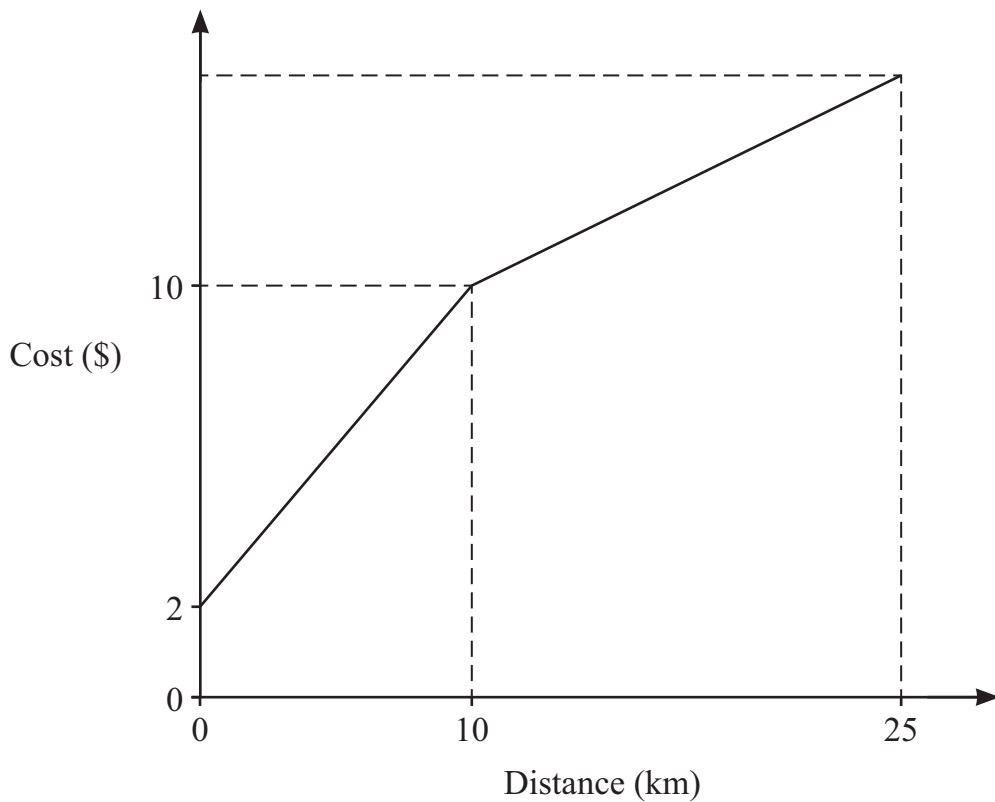
plus

- \$ x per km for the first 10 km of a journey

plus

- \$0.50 per km for the next 15 km of a journey.

The graph shows information about these costs.



- (a)** Find the rate, \$ x per km, for the first 10 km of a journey.
Use the graph to help you.

$x =$ [1]

- (b)** Find the total cost for a journey of 25 km.
Use the graph to help you.

\$ [1]

- 19 The scale drawing shows the positions of two towns, A and B .

Scale 1 : 50 000



Town C is 3 km from town B .

The bearing of town C from town A is 040° .

Draw the **two** possible positions of town C .

[3]

20 Complete the equations by writing an integer power in each box.

$$7^{-2} \div 7^{\boxed{}} = 7^{-6}$$

$$(9^2)^{\boxed{}} = 9 \div 9^7$$

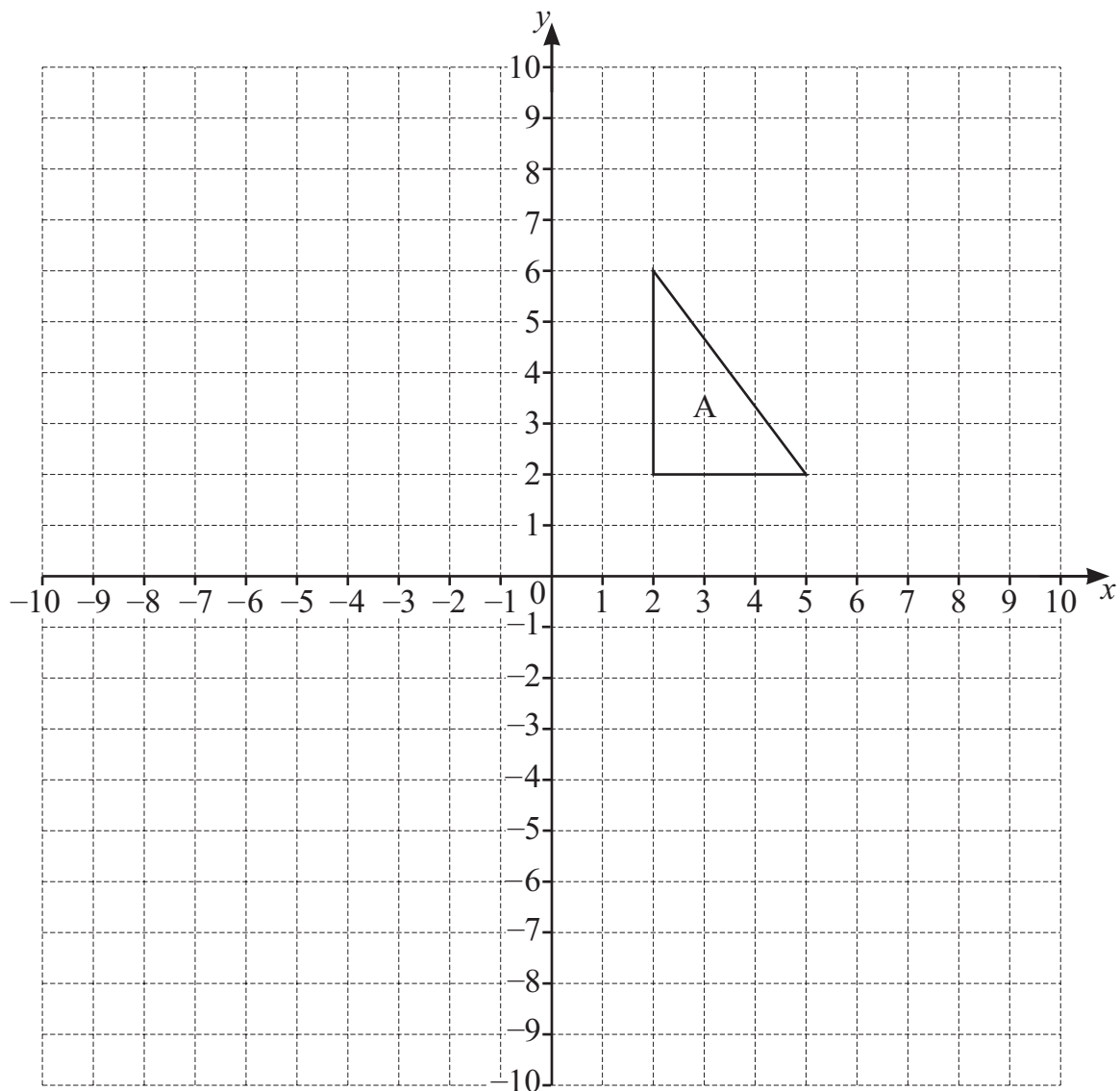
[2]

21 A rectangle has a width of $4x$ cm and an area of $(16x^2y + 4x)\text{cm}^2$

Find an expression, in terms of x and y , for the length of the rectangle.

..... cm [2]

22 Shape A is shown on the grid.



Shape A is mapped onto shape B by a reflection in the line $x = 1$

Shape B is mapped onto shape C by a reflection in the line $y = -2$

Describe fully the **single** transformation that maps shape A onto shape C.
You may use the grid to help you.

.....
..... [2]

23 Work out.

$$2\frac{1}{5} - \frac{14}{15} \times \frac{3}{8}$$

Give your answer as a mixed number in its simplest form.

..... [3]

24 Ahmed thinks of a number.

He adds 12 to his number and then divides by 2

Safia thinks of a number.

She multiplies her number by $\frac{3}{2}$ and then takes away 4

Ahmed says,

‘There is one number we could both think of that would give us the same answer.’

By writing and solving an equation in terms of n , find the answer that Ahmed and Safia both get.

Answer = [4]

25 Point P has coordinates $(p, 4)$.

Point Q has coordinates $(q, 4)$ where $q < 0$

Q is the image of P after a reflection in the line $x = -5$

Mia says, ' p **must** be greater than -5 '.

Find a possible value of p to show that Mia is **not** correct.

$p = \dots\dots\dots$ [1]

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